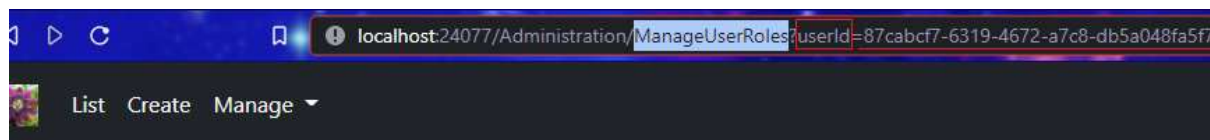


## 92 - Model binding not working on submitting razor view with foreach loop

Защо присвояването на данни чрез URL не работи адекватно с for each цикъл:



### Manage User Roles

- ☐ User
- ☐ Admin
- ☐ TestRole1

Update

Cancel

```
AdministrationController.cs  X
- EmployeeManagement.Controllers.AdministrationC - ManageUserRoles(List<UserRo

[HttpGet]
0 references
public async Task<IActionResult> ManageUserRoles(string userId)
{
    ViewBag.userId = userId;

    var user = await userManager.FindByIdAsync(userId);

    if (user == null)
    {
        ViewBag.ErrorMessage = $"User with Id = {userId} cannot be found";
        return View("NotFound");
    }
}
```

Накратко: Get приема от URL userId -> създава променлива model, която е list и попълвайки всички нейни свойства, всеки път създава нов обект в инстанция, който после добавя към list и предава на View.

```

var model = new List<UserRolesViewModel>();

foreach (var role in roleManager.Roles)
{
    var userRolesViewModel = new UserRolesViewModel
    {
        RoleId = role.Id,
        RoleName = role.Name
    };
    // populate properties Id, Name, IsSelected
    if (await userManager.IsInRoleAsync(user, role.Name))
    {
        userRolesViewModel.IsSelected = true;
    }
    else
    {
        userRolesViewModel.IsSelected = false;
    }

    model.Add(userRolesViewModel); // add to list
}
// -> View(model);

```

Очаквано, моделът, който подаваме на View е list, като най-важното е, че цикъл for за всеки обект от списъка кара razor да генерира тези групи елементи (2 скрити, един чек-бокс и 1 лейбъл за името), като submit btn Update потвърждава и изпраща формата до post.

```
controller.cs ManageUserRoles.cshtml X
@model List<UserRolesViewModel>
@{
    var userId = ViewBag.userId;
}

<form method="post">
    <div class="card">
        <div class="card-header">
            <h2>Manage User Roles</h2>
        </div>
        <div class="card-body">
            @for (int i = 0; i < Model.Count; i++)
            {
                for each object in this list Razor generate this set of input elements
                <div class="form-check m-1">
                    <input type="hidden" asp-for="@Model[i].RoleId" />
                    <input type="hidden" asp-for="@Model[i].RoleName" />
                    <input asp-for="@Model[i].IsSelected" class="form-check-input" />
                    <label class="form-check-label" asp-for="@Model[i].IsSelected">
                        @Model[i].RoleName
                    </label>
                </div>
            }
            <div asp-validation-summary="All" class="text-danger"></div>
        </div>
        <div class="card-footer">
            <input type="submit" value="Update" class="btn btn-primary"
                style="width:auto" />
            <a asp-action="EditUser" asp-route-id="@userId"
                class="btn btn-primary" style="width:auto">Cancel</a>
        </div>
    </div>
</form>
```

- ☐ User
- ☐ Admin
- ☐ TestRole1

Post методът автоматично ще присвои на model тип list 3-те input елемента (2 hidden + 1 check-box isSelected)

```
AdministrationController.cs  X
EmployeeManagement.Controllers.AdministrationC  ManageUserRoles(List<UserRole

[HttpPost]
public async Task<IActionResult>
ManageUserRoles(List<UserRolesViewModel> model, string userId)
{
    var user = await userManager.FindByIdAsync(userId);

    if (user == null)
    {
        ViewBag.ErrorMessage = $"User with Id = {userId} cannot be found";
        return View("NotFound");
    }

    var roles = await userManager.GetRolesAsync(user);
    var result = await userManager.RemoveFromRolesAsync(user, roles);

    if (!result.Succeeded)
    {
        ModelState.AddModelError("", "Cannot remove user existing roles");
        return View(model);
    }

    result = await userManager.AddToRolesAsync(user,
        model.Where(x => x.IsSelected).Select(y => y.RoleName));
}
```

```
if (!result.Succeeded)
{
    ModelState.AddModelError("", "Cannot add selected roles to user");
    return View(model);
}

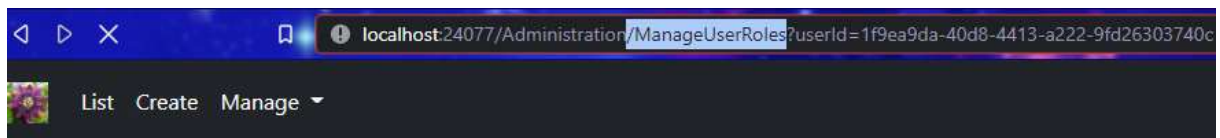
return RedirectToAction("EditUser", new { Id = userId });
}
```

За да разберем това автоматично присвояване на 3-те input -> Post, поставяме Breakpoint:

```
ManageUserRoles.cshtml*  AdministrationController.cs*  X
EmployeeManagement  EmployeeManagement.Controllers.AdministrationC  ManageUserRoles

171  [HttpPost]
172  public async Task<IActionResult>
173  ManageUserRoles(List<UserRolesViewModel> model, string userId)
174  {
175  var user = await userManager.FindByIdAsync(userId);
176
```

Debug ->

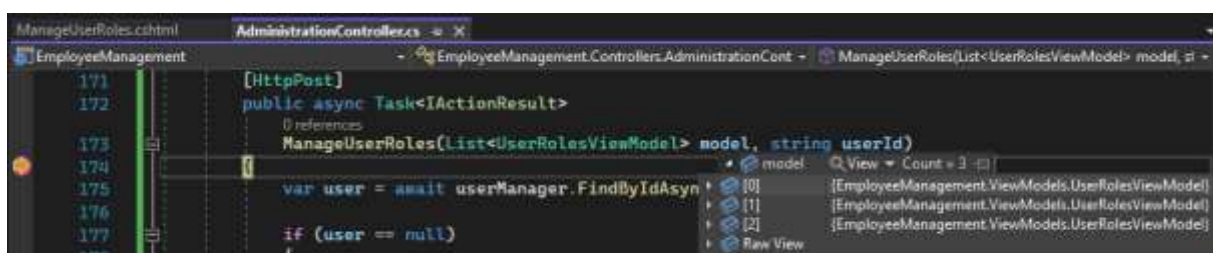


## Manage User Roles

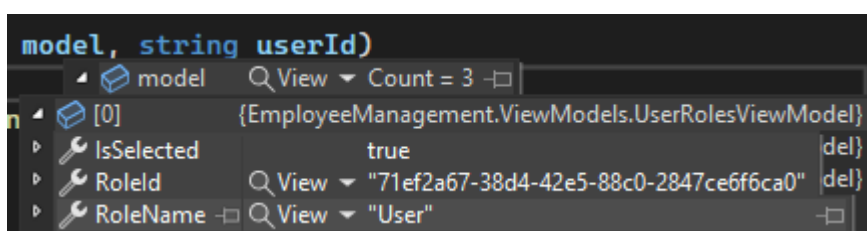
- ☒ User
- ☒ Admin
- ☒ TestRole1

```
<div class="card-header">
  <h2>Manage User Roles</h2>
</div>
<div class="card-body">
  <div class="form-check m-1">
    <input type="hidden" id="z0_RoleId" name="[0].RoleId" value="71ef2a67-38d4-42e5-88c0-2847ce6f6ca0" />
    <input type="hidden" id="z0_RoleName" name="[0].RoleName" value="User" />
    <input class="form-check-input" type="checkbox" data-val="true" data-val-required="The IsSelected field
    <label class="form-check-label" for="z0_IsSelected">
      User
    </label>
  </div>
  <div class="form-check m-1">
    <input type="hidden" id="z1_RoleId" name="[1].RoleId" value="7a53c902-d9f9-4026-b521-26b8485f1611" />
    <input type="hidden" id="z1_RoleName" name="[1].RoleName" value="Admin" />
    <input class="form-check-input" type="checkbox" checked="checked" data-val="true" data-val-required="Th
    <label class="form-check-label" for="z1_IsSelected">
      Admin
    </label>
  </div>
  <div class="form-check m-1">
    <input type="hidden" id="z2_RoleId" name="[2].RoleId" value="e51deba4-def5-46fa-9acd-b9a75e3a1259" />
    <input type="hidden" id="z2_RoleName" name="[2].RoleName" value="TestRole1" />
    <input class="form-check-input" type="checkbox" checked="checked" data-val="true" data-val-required="Th
    <label class="form-check-label" for="z2_IsSelected">
```

Update -> BreakPoint, която вече е hit с жълта стрелка и ако селектираме model, той вече е възприел 3 елемента:



и ако изберем всеки от тях, той съдържа попълнени параметри, автоматично прихванати от Input таговете във формата. Това е благодарение на model-binding MVC.



Continue -> само че при мен не позволява да продължа без редакция на кода, за разлика от по-старата версия. Явно трябва да вида в настройките.

Спираме Debug и вместо for, нека видим при foreach:

```
@for (int i = 0; i < Model.Count; i++)
```

```
@for (int i = 0; i < Model.Count; i++)
```

```
@foreach (var role in Model )
```

```
@foreach (var role in Model )
```

Debug -> View source -> резултатът е потресаващ: независимо кой label с име или check-box се селектира, само първият елемент се маркира (в моя браузър не е така), но при ViewSource:

Стара версия (дублиране на hidden id = name, checkbox id = for)

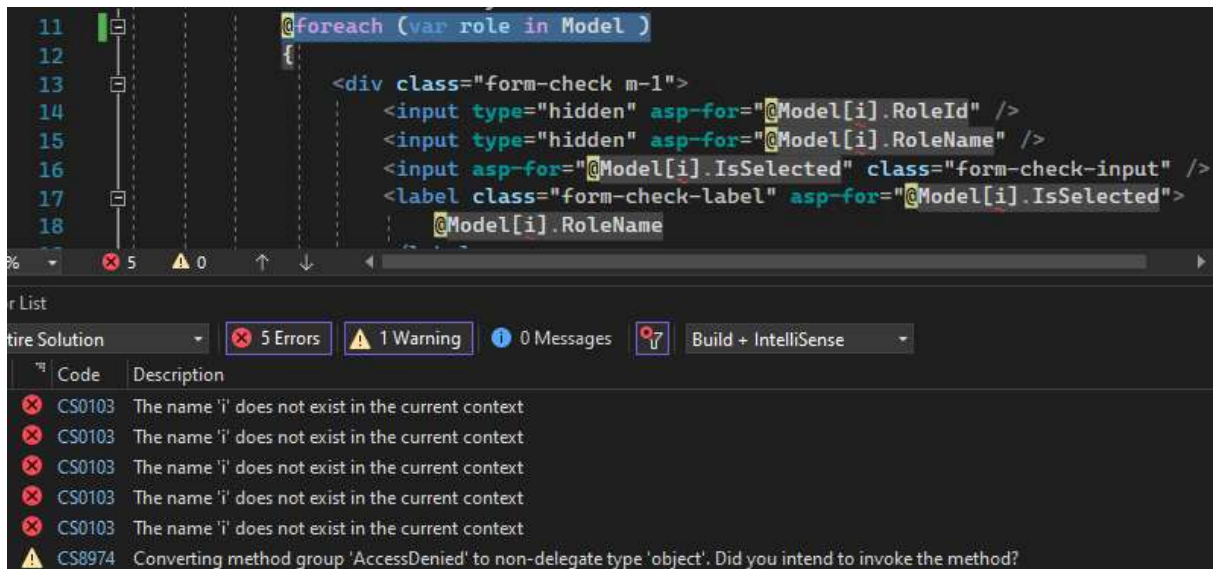


Нова (дублиране само на checkbox id = for):

```
<div class="form-check m-1">
  <input type="hidden" id="r0_RoleId" name="[0].RoleId" value="71ef2a67-36d4-42e5-80c0-2047ce9f8ca0" />
  <input type="hidden" id="r0_RoleName" name="[0].RoleName" value="User" />
  <input class="form-check-input" type="checkbox" checked="checked" data-val="true" data-val-required="The IsSelected field is required." id="r0_IsSelected" name="r0_IsSelected" />
  <label class="form-check-label" for="r0_IsSelected">
    User
  </label>
</div>
<div class="form-check m-1">
  <input type="hidden" id="r1_RoleId" name="[1].RoleId" value="7a55c902-b9f9-4028-b121-2608485f1611" />
  <input type="hidden" id="r1_RoleName" name="[1].RoleName" value="Admin" />
  <input class="form-check-input" type="checkbox" checked="checked" data-val="true" data-val-required="The IsSelected field is required." id="r1_IsSelected" name="r1_IsSelected" />
  <label class="form-check-label" for="r1_IsSelected">
    Admin
  </label>
</div>
<div class="form-check m-1">
  <input type="hidden" id="r2_RoleId" name="[2].RoleId" value="e51deba4-def5-40fa-8acd-b9a75e3a1259" />
  <input type="hidden" id="r2_RoleName" name="[2].RoleName" value="TestRole1" />
  <input class="form-check-input" type="checkbox" data-val="true" data-val-required="The IsSelected field is required." id="r2_IsSelected" name="r2_IsSelected" />
  <label class="form-check-label" for="r2_IsSelected">
```

Виждаме, че има проблем: присвоените стойности на id = for и за трите елемента се дублират, което се отчита с множество грешки и червени подчертавания на i. По същият начин пада на

първия и втория hidden input съвпадат. Поради всичко това, model binder of MVC не знае кои от тези входни параметри къде да присвои.



```
11  @foreach (var role in Model)
12  {
13      <div class="form-check m-1">
14          <input type="hidden" asp-for="@Model[i].RoleId" />
15          <input type="hidden" asp-for="@Model[i].RoleName" />
16          <input asp-for="@Model[i].IsSelected" class="form-check-input" />
17          <label class="form-check-label" asp-for="@Model[i].IsSelected">
18              @Model[i].RoleName
19      </div>
20  }
```

5 Errors 1 Warning 0 Messages

| Code   | Description                                                                                                |
|--------|------------------------------------------------------------------------------------------------------------|
| CS0103 | The name 'i' does not exist in the current context                                                         |
| CS0103 | The name 'i' does not exist in the current context                                                         |
| CS0103 | The name 'i' does not exist in the current context                                                         |
| CS0103 | The name 'i' does not exist in the current context                                                         |
| CS0103 | The name 'i' does not exist in the current context                                                         |
| CS8974 | Converting method group 'AccessDenied' to non-delegate type 'object'. Did you intend to invoke the method? |

Затова, когато направим промяна, натиснем Update, BreakPoint is hit, посочим model, в старите версии няма присвояване, а model count = 0. При мен обаче, промените са отразени, което е много добре. Явно затова ми е позволявало маркиране на всички check-box, не само първия.

## Model Binding Not Working

- With a FOR loop, the generated input element names are unique and contains an integer indexer in the name
- So the model binder knows it has to map the values from these input elements to the list parameter on the controller action
- With a FOREACH loop the names are not unique and hence the model binder does not know it has to map the values to the list parameter

Възстановявам for.